



Accessible Real-time Eye-gaze Tracking for Neurocognitive Health Assessment: A Multimodal Web-based Approach

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Abstract

We introduce a novel integration of real-time predictive eye-gaze tracking models into a scalable cloud-based multimodal dialogue system tailored for remote health assessments. A virtual human guide, Tina, interacted with 10 participants with Mild Cognitive Impairment (MCI) and 29 healthy controls during a crowdsourced pilot study. She engaged participants in an approximately 10-minute interview with 11 interactive eye-gaze tasks, ranging from simple free gaze exploration to more specialized directed gaze tasks. We found that metrics of eye-gaze dynamics and reaction times extracted from these tasks, combined with an Adaboost classifier effectively differentiates MCI participants from healthy adults, with an average accuracy of 0.94. Furthermore, over 80% of participants reported high engagement and liked the user experience with the platform, across demographics and internet connections, demonstrating its robustness and scalability for real world use.

Index Terms: eye-gaze tracking, remote health monitoring, multimodal health assessments, Mild Cognitive Impairment, Mild Neurocognitive Disorder

1. Introduction

Eye-gaze tracking has emerged as a valuable method for detecting and monitoring neurological disorders due to the close anatomical proximity between the optical nerve and the brain [1]. However, existing research predominantly relies on expensive, specialized equipment within tightly controlled settings, rendering these standards impractical and unsustainable for everyday health tracking. To address this, recent advancements in remote video technology, accelerated by the COVID-19 pandemic, have made high-quality webcam feeds more affordable and ubiquitous. In this study, we explore the feasibility of using conventional computer and laptop webcams for eye-gaze tracking in a remote healthcare setting, aiming to bridge the gap between clinical precision and practicality.

Additionally, breakthroughs in real-time machine learning within web browsers have enabled real-time gaze tracking. Coupled with developments in multimodal dialogue agents for healthcare applications, an exciting opportunity arises: a multimodal dialogue system that incorporates eye-gaze tracking for health assessments. Leveraging clinically proven oculomotor exercises adapted as real-time browser tasks, we conduct a crowdsourced patient study involving individuals with Mild Cognitive Impairment (MCI) and healthy controls. By analyzing patients' gaze patterns during interactions with our augmented multimodal callflow, we extract valuable features and metrics for health assessment. Our system, designed to assist remote and accessible health assessments, holds promise not only for MCI but also as a potential approach for other neuro-

logical disorders. Through this innovative approach, we aim to alleviate barriers to identification and tracking, ultimately facilitating thorough screenings and specialized treatment plans for patients with neurological conditions.

2. Related Work

2.1. Conversational Remote Health Assessment

We leverage and build upon the Modality cloud-based conversational platform for remote patient assessment and monitoring [2, 3], the utility which has been demonstrated across multiple neurological and mental health disorders, including ALS [4], Parkinson's disease [5], autism [6], schizophrenia [7], depression [8] and MCI [9]. A friendly virtual human, Tina, guides participants through a conversational interaction comprising several speaking and cognitive exercises, during which their speech and video data is recorded and relevant biomarkers extracted therefrom. These biomarkers can be visualized on an intuitive dashboard for in-depth analysis. The conversational protocol can be adapted to the specific neurological condition being assessed, incorporating tasks such as oral motor examinations, speech assessments, fine motor exercises and memory recall tasks.

2.2. Eye-gaze Tracking for Healthcare

The medial longitudinal fasciculus, a critical neuronal structure coordinating eye movements, is susceptible to early manifestations of neurocognitive disorders. Eye tracking holds promise for identifying and monitoring such conditions. Historically, eye tracking research has relied on specialized hardware in controlled environments. Studies, including [10], demonstrate eye tracking's accuracy in distinguishing Parkinson's disease from controls and assessing ADHD and fetal alcohol spectrum disorder. [1] and [11] further explore eye-gaze features, emphasizing their potential as non-invasive biomarkers for cognitive health.

More recently, WebGazer [12, 13], an online eye-gaze tracker, leverages Tensorflow.js and Facemesh for real-time gaze estimation using webcam feeds without the aid of specialized equipment [14, 15]. Additionally, smartphone-based eye-gaze tracking, as explored by Valliappan et al. (2020), demonstrates high precision (mean error of approximately 0.50 ± 0.03 cm in-person on a phone screen) despite not being real-time. The ongoing pursuit of real-time capabilities marks a significant milestone in democratizing eye-gaze tracking for accessibility [16]. Leveraging WebGazer's robust pretrained model, our gaze estimation system prioritizes frontend-based architecture, webcam tracking, and ease of browser development.

3. System

To create an environment for online eye tracking for health assessments, we adapted and integrated WebGazer into our existing multimodal dialog platform. Although Webgazer excels at live gaze estimations and can be deployed on the frontend of browsers through TensorflowJS, it relies on the location of a participants' mouse cursor to estimate the most probable gaze point on screen [12, 13]. We replicated and matched the error rates of the original WebGazer paper at approximately 180 px while running concurrently with the multimodal dialogue system. The error rates along the x-axis being notably less than on the y-axis (x-axis error upper-bound: 11%, y-axis upper-bound: 21%). These error rates informed many user-interface design decisions involving size and spacing, with most objects being set at least 200 pixels apart and images being displayed at a large enough size to where points of interest are approximately 200 pixels apart. Other key findings being carried into this work is the need to include a manual calibration stage at the beginning of calls where the user is asked to click circles around the screen for a certain number of times each while looking directly at the circle they are clicking. Following this step, we block cursor data to the WebGazer model to allow for higher accessibility while engaging participants in interactive tasks guided by audio prompts from the conversational agent. This combination of calibration and continued interaction allows eye-gaze predictions to maintain accuracy over long call durations.

We chose tasks from the literature that have been demonstrated to elicit eye-gaze behavior (and corresponding metrics) that are discriminative of neurological disease. The tasks, in order of presentation to the participant, are as follows:

1. **Calibration Task.** Users find and click buttons on the edge of the browser window (this helps calibrate relative eye positions to the limits of the edges of the screen). The last 500 frames of data are reported as one segment to maintain uniformity between calls and minimize data usage.
2. **Extreme Saccade Task.** Users begin with exaggerated looks as far left and right as possible, then are prompted to repeat the movements of looking back and forth as quickly as they can while keeping their head still (no stimulus on screen for this task). The task is reported as one continuous segment.
3. **Smooth Pursuit Task.** Users follow a small circle on screen moving smoothly and slowly in a predetermined path (horizontal line and circular path) across the screen using gaze. In the *Horizontal Line* subtask, starting from the center, the circle moves to the left edge of the screen, moves to the right edge, then returns to the center to complete a cycle. The task consists of 3 cycles, with each one being recorded as a separate subtask. The *Circle* subtask (see figure 1a) starts above the center of the screen at a 12 o'clock location and follows a clockwise path around the center of the screen, completing a revolution when reaching its original position. The task consists of 3 with revolutions, each one being recorded as a separate subtask.
4. **Saccade Task.** Users find and hold their gaze on objects as they appear for 3 seconds, disappear, and appear at different positions on screen. The locations of the objects appear random, but are predetermined and in a set order to maintain uniformity between calls. Each saccade to a new object on screen, 8 in total, are recorded separately.
5. **Stroop Task.** In the *Congruent Stroop* variant, users read a list of colors in which the text and their colors match within a certain time limit. In the *Incongruent Stroop* variant, users

read a list of colors in which the text and their colors are mismatched within a certain time limit (see 1b).

6. **Image Exploration Task.** In the *Directed Image Exploration* subtask, users are prompted to find and hold their gaze on 3 different objects within the image under a certain time limit, one at a time. Each search and saccade attempt is recorded separately. In the *Free Image Exploration* version, users are prompted to freely scan and look at an image for a period of time (see figure 1c).

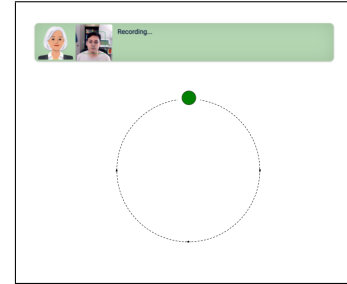
4. Methods

4.1. Data

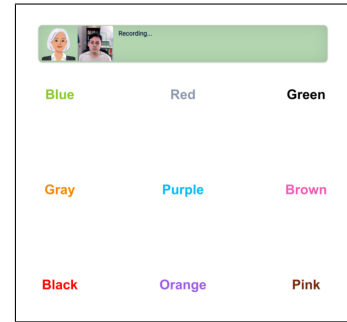
We conducted a crowdsourced observational study in which healthy controls and patients, as determined by self-reports their neurological health, engaged with our system.

Sessions	Control	MCI	All
Partial	29	10	39
Complete	27	8	35

Table 1: Crowdsourced Study Population



(a) Smooth Pursuit Circle Task Example



(b) Incongruent Stroop Task Example



(c) Image Exploration Task Example

Figure 1: Examples of 3 assessment protocol tasks

Metric	Number of Subtasks	Total Metrics
Displacement	20	180
Velocity	20	180
Acceleration	20	180
Distance from Stimulus	13	130
Time to Target	13	117
# of Times on Target	13	39
Total Time on Target	13	117
Overall	112	943

Table 2: Overview of metrics computed from Euclidean coordinates of estimated eye gaze, including mean, median, max and standard deviation statistics (for tasks involving a specific stimulus/target, metrics were computed against that target only).

4.2. Preprocessing

The data stream obtained after study completion—comprising raw video, facial landmarks, audio, survey demographics, and gaze data—was separated to focus exclusively on the analysis of gaze data. To streamline the dataset, we combined all subtasks into their respective task, preserving the flexibility to examine individual subtasks for future analysis if deemed necessary. We then labeled each participant and documented the number of tasks they completed. Noting that with the subtask structure of the callflow, we can use incomplete calls for training and testing as long as we have collected a certain amount of eye-gaze data from the sessions.

To ensure consistency across the dataset, and accounting for variations in user window sizes and potential resizing during tasks, we employed a the following normalization process for the predicted gaze data. This approach was pivotal in unifying the data, thereby facilitating a robust analysis of gaze patterns across diverse user interactions.

- Fitting:** The maximum window width and height for that task are taken, and all points are treated to be within the limits of these dimensions.
- Normalizing:** Using the maximum window dimensions during that task, all gaze points are converted to be between [0, 1] for both the x-axis and y-axis.
- Smoothing:** All points are then run through a Savitzky-Golay (Savgol) filter with a window of 3 points and a polynomial order of 2. This helps eliminate extreme outliers while maintaining jittery movements which can be exhibited by users with motor impairments.
- Saccade Segmentation:** For tasks with stimuli (even those not designed for saccade movements), windows of gaze history in which the predicted gaze moves towards and stays on a stimulus (Time to target and Time on target) are segmented for additional metrics.

4.3. Metrics

Seeking to create robust and adaptable gaze tracking metrics, we aimed to develop generic measures that could be applied across a wide range of healthcare applications, extending beyond specific task or timing constraints. While deep learning models such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), or Transformers are well-suited for time series data, our focus is towards human-interpretable classifications using quantitative data. Given the scarcity of data and the desire for transparent insights, we opted to derive metrics from easily understandable features, avoiding the complexity of “black box” deep learning models (although we acknowledge their potential for future enhancements).

As part of our generic metric suite applicable to all tasks, we computed velocities and accelerations for the x and y axes, along with absolute Euclidean values of velocities and accelerations for the predicted gaze points. Specifically, we analyzed the mean, median, maximum, and standard deviation of these metrics. For tasks involving targets (such as gaze fixation on specific points), we extended our analysis to include x, y, and Euclidean distances from the target. Again, we leveraged the mean, median, maximum, and standard deviation to summarize these distance metrics. Furthermore, we quantified the mean squared errors for each list of targets, providing a comprehensive assessment of gaze prediction accuracy. Finally, to capture temporal aspects and saccade-related metrics, we devised specific measures applicable to all tasks with targets, such as: Time to target, Time on target, Times on target. Note that with these generic metrics, each user has the number of tasks times the number of metrics, resulting in a large number of features compared to the population.

5. Analyses and Results

5.1. Prediction Error vs. Errors from MCI

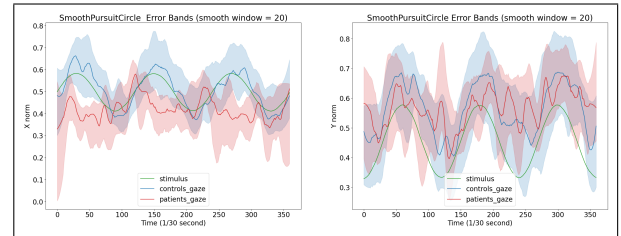


Figure 2: Average Gaze vs. Stimulus Location (Smooth Pursuit Circle Task)

Using the smooth pursuit tasks as ground truth for gaze location of healthy controls in the crowdsourced study, we were able to estimate upper-bounds for our gaze prediction error to be at 14.7% on the x-axis and 18.3% on the y-axis among healthy controls. Users with MCI had an observed estimated error of 15.3% for the x-axis and 19.3% for the y-axis under the same methodology (average window size: 1332 x 820 px). Although the error between the cohorts were similar, visualizations of the tested tasks indicate distinct trends, as seen in fig. 2 with MCI participants’ being off phase of the revolving circle. This and similar insights underscore the need for a nuanced understanding of the sources of error in gaze prediction.

5.2. Feature Selection

The significant number of features in comparison to the population size necessitated dimensionality reduction. To address this, we explored different feature selection methods. Initially, we employed Principal Component Analysis (PCA) [17]. The linearization of the data emerged as an effective exploratory step, streamlining features for subsequent classification tasks. However, as our accuracy levels improved, we encountered limitations associated with PCA. Notably, PCA assumes that all dimensions are independent, which can lead to overfitting issues when dealing with related features. One potential solution is parametric bootstrapping, although this approach introduces scalability concerns for future applications [18]. Additionally, alternative techniques like Principal Component Regression (PCR) allow for further analysis and reduction of components [19]. Despite these options, the fundamental inter-

pretability challenge posed by PCA components remains, as their analysis lacks intuitive clarity.

To improve the interpretability of our data, we posit that the mean and distribution shape of gaze features between cohorts are different (and potentially non-normal), potentially helping with each cohort’s respective categorization. But there are limitations in using a non-ordinal and categorical approach to the data. Although this approach allows for further expansion into examining additional disorders and/or impairments, it limits the scope of future research by collapsing the wide spectrum of NCD patients into discrete, non-ordinal groups.

To approach this hypothesis, we perform Kruskal-Wallis tests, with the null hypothesis being that the mean of the distributions between cohorts are the same [20]. This analytical process yields p-values for each feature, outlining the statistical significance of these features in deviating from the null hypothesis and revealing distinctive features that exhibit different distributions across cohorts. Using these p-values, we group and observe the most distinctive features under this methodology. Notably, the most characteristic metrics (p-value ≤ 0.01 ; 13 features) were max and standard deviations of distances, velocities, and accelerations; the mean and median values of the metrics being more moderately significant (p-value ≤ 0.05 ; 30 features) with time on target and time to target also being included in this group. Similar to the correlation ranking approach, the most indicative tasks were saccade-based ones but with a higher skew towards free exploration tasks and with the smooth pursuit tasks also being moderately significant as well.

5.3. Classification

After dimensional reduction, feature ranking, and p-value grouping, we performed multiple methods of binary classification on various subsets of the features generated using the above approaches. We further used Kruskal-Wallis tests to select features that demonstrated statistically-significant differences between MCI and healthy controls. We used the following classifiers: Baseline (majority vote), Nearest Neighbors, Linear SVM, RBF SVM, Gaussian Process, Decision Tree, Random Forest, Neural Net, AdaBoost, and Naive Bayes [21]. Due to the small sample size of the study, we measured the average accuracy over k-fold (k = smallest class size) cross validation. The best performance for PCA was on completed sessions using an AdaBoost classifier with 50-estimators with an accuracy of 0.89 (F1 Score = 0.67) over a baseline of 0.77.

	Best Classifier	Max p-val	Baseline Accuracy	Average Accuracy	Average F1
All Sessions	Naive Bayes	0.01	0.74	0.88	0.70
Completed Sessions	AdaBoost	0.01	0.77	0.94	0.85

Table 3: *Best performing classifier (after Kruskal-Wallis feature selection)*

Our experiments suggest that AdaBoost classifiers perform best using these features. Although not as intuitive as many other classifiers tested, AdaBoost classifiers can be “human understandable” when decomposed into its subcomponents of small trees or “stumps.”

Last but not least, qualitative post-call surveys 3 demonstrated that 82% of participants reported both high engagement and high performance (rating ≥ 4 on a 5-point Likert scale from 1-Least Satisfactory to 5-Most Satisfactory) while interacting

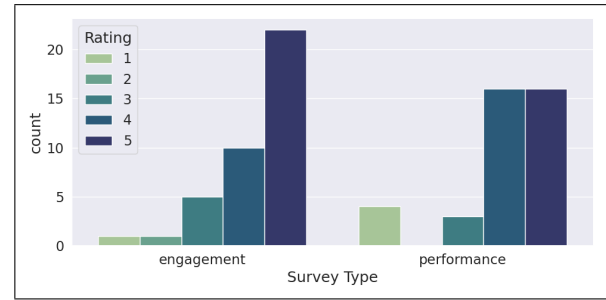


Figure 3: *Results of post-call usability surveys (N = 39; higher ratings reflects better engagement and performance levels)*

with the system. This is important as in addition to high performance, a robust user-experience is a crucial factor impacting the design and practical adoption of assessment systems in the real world.

6. Discussion

This paper introduces the first end-to-end multimodal dialog system that employs conventional webcam video input for eye-tracking to facilitate remote health assessments via the web. Our implementation seamlessly integrates predictive eye-gaze as a novel modality within a cloud-based remote multimodal health system, and is robust, cost-effective and highly scalable. By capturing synchronized data from various user input modes, we break down information efficiently and implement safeguards against errors or missing data. Notably, this synchronized data not only allows for extended multimodal fusion models, future re-analysis through call recreation, but also holds promise for training new eye-gaze models using webcam data – an important avenue for future work towards improving existing model performance. Minimizing cursor usage and relying on machine-translated audio prompts enhance accessibility and enable consistent monitoring for longitudinal studies, which could offer deeper insights into conditions’ etiology and progression. We observed the benefits of these engineering and software architecture choices reflected in user experience (UX) surveys of our participants, who overwhelmingly rated the system as being engaging and performing well overall. Finally, we demonstrated that a comprehensive set of eye gaze tracking features computed using such a system produced excellent classification performance at discriminating patients with mild cognitive impairment (MCI) from healthy controls. Importantly, our research prioritizes interpretability, ensuring healthcare professionals intuitively grasp eye-gaze patterns’ neurocognitive implications.

That being said, there are some limitations to the current work that provide avenues for future exploration. For example, we did not consider deep learning architectures for our classification experiments, for two reasons. The first is the relatively small sample size of the MCI cohort. While small sample sizes are typical of such studies in the healthcare domain, they are nonetheless insufficient to train competitive deep learning models. Future work will investigate the performance of this system on larger sample sizes and a more diverse array of neurological, neurocognitive and neurodevelopmental disorders. The second is interpretability, as mentioned earlier – today’s deep learning models are still largely black boxes, and healthcare domains require models that are explainable. That being said, there is plenty of ongoing research on making these models more explainable, and future research will investigate how we can leverage the power of such methods to push classification accuracies without compromising clinical interpretability.

7. References

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